

MUHAMMED HASAN

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Education

Siirt University

Bachelor of Engineering in Mechanical Engineering

Sep 2021 – May 2025

Siirt, Turkey

Experience

Deneyap Technology Workshop

Instructor & Mentor

Mar 2024 – Jan 2026

Siirt, Turkey

- Instructed 100+ students in robotics, embedded programming, and renewable energy systems
- Guided 5 student teams through 6-month product development cycles, with all teams delivering prototypes on schedule

TUBITAK 2209A Research Program

Undergraduate Researcher | Supervisor: Dr. M. Rasit Atelge

Mar 2024 – Mar 2025

Siirt, Turkey

- Synthesized supercapacitor electrodes from agricultural waste achieving 7.98 F/g specific capacitance
- Reduced experimental trials by 67% via Taguchi optimization, compressing an 8-month research plan to 3 months
- Trained ANN and RF models in Python achieving 94% prediction accuracy across 50+ material characterization tests
- Performed electrochemical characterization via cyclic voltammetry, galvanostatic charge-discharge, and EIS

Clean Energy Laboratory, Siirt University

Research Intern | Supervisor: Dr. Fatih Bayrak

Jun 2024 – Aug 2024

Siirt, Turkey

- Designed instrumentation setup collecting thermal data across 18 PV spray-cooling configurations, contributing experimental results to an Applied Thermal Engineering publication
- Achieved 15°C temperature reduction and 26% power increase, validating 47% peak cooling drop.

Projects

Autonomous Fixed-Wing UAV | Mechanical Design

Mar 2026 – Present

- Designing 1.7m fixed-wing airframe with XPS core, carbon spars, and vacuum-infused composite skin
- Validating structural integrity via hand calculations and mechanics equations under aerodynamic and inertial loads
- Optimizing mass distribution and component placement to maintain the CG at 28% MAC for flight stability

HydroFinance AI | Project Developer

Jan 2026 – Present

- Develops an AI-driven techno-economic SaaS platform automating LCOH, NPV, IRR calculations for green hydrogen.
- Engineers a Python backend integrating ERA5 weather data to simulate hourly electrolyzer physics and grid dispatch.

Annular Combustion Chamber Design | Team Lead

Dec 2024 – Apr 2025

- Directed nine-member team engineering annular combustion chamber for 60 lbf turbojet engine
- Coordinated combustion physics modeling, CFD analysis, materials selection, and additive manufacturing workflows

Quantum-Enhanced Solar Forecasting | Project Developer

Jun 2024 – Jan 2025

- Built hybrid quantum-classical neural network using PennyLane for solar irradiance prediction.
- Benchmarked against classical ML (SVR, RF, XGBoost) achieving similar accuracy with 40% faster training.

Thorium Molten Salt Reactor Design | 6th Place (100+ Teams)

Jul 2024 – Oct 2024

- Designed a 135 MWe reactor for the Nuclear Energy Competition, achieving 42% thermal efficiency
- Engineered supercritical CO₂ Brayton cycle cooling configuration balancing heat rejection against parasitic losses

Battery Thermal Management System | Team Lead, Finalist – Top 10 of 171 Teams

Nov 2023 – Sep 2024

- Led a 4-member team designing a microchannel cooling system with CuO nanofluid for lithium-ion batteries, achieving 31% temperature reduction in 18650 cells at 2C discharge with $\pm 3^{\circ}\text{C}$ uniformity
- Built and validated a working prototype targeting integration into 50+ kWh automotive battery packs

Biomass Supercapacitor | Team Lead, 3rd Place of 50+ Teams Globally

Jun 2023 – Dec 2023

- Fabricated biomass-derived electrodes reaching 280 F/g, matching commercial activated carbon performance
- Integrated the supercapacitor with dye-sensitized solar cells into a renewable energy storage demonstration system

Technical Skills

Simulation & Analysis: ANSYS Fluent, CFD, FEA, MATLAB, Python, HOMER Pro, Electrochemical Analysis

Design & Manufacturing: SolidWorks, Fusion 360, GD&T, Additive Manufacturing, Composite Fabrication

Languages: English (C1), Turkish (C1), Arabic (Native), Kurdish (Native)

Publications

2x Conference Papers – Bogazici Scientific Research Congress & Sivas International Conference, 2025

1x Poster Presentation – Batman University, 2024